Port Command: Prototype Specification

Oct 28, 2026 · @Mahian Ferdous

Purpose and scope

The prototype is a front-end-only demo of one simulated port, built to show in five minutes that an AI layer can run a port; it needs no real backend or database.

Who it is for. Competition judges first, then the MMC Group pitch. It must look like a control-room product used every day, never like a game. The matching pitch is in the pitch document.

Working assumptions (change any of them, but tell the pitch author):

- Runs in a browser on one laptop, full screen at 1920 by 1080, with no internet needed on stage.
- One fully modelled fictional terminal, called Selat Terminal here, with 4 berths, 3 vessels, about 40 vehicles, about 60 worker pins and about 2,000 containers.
- The AI copilot answers a fixed set of questions deterministically; a live language model is optional.

Part	Real in the prototype	Simulated or faked
Map, animation, zoom, inspectors	Yes	Nothing
Vessels, vehicles, workers, containers	The motion and the interaction	All positions and records
Alerts and safety rules	The rule logic, such as a zone breach	The incidents
AI copilot	The interface and the proposal flow	The answers, unless a live model is wired in
Portals	The screens	Their data
Login, database, real integrations	Not built	Not applicable

Non-negotiables

1. Professional look: restrained palette, no neon, no game-style effects.
2. Smooth motion at 60 frames per second on the presenting laptop.

3. Every click in the stage script works, and every scene can be replayed.
4. A visible "Simulated data" badge on every screen.

How it compares with existing systems

Port Command does not compete with a terminal operating system or a crane control stack; it sits above them as an AI decision layer and a role-based front end. Product names are examples, not endorsements; check vendor details before quoting them.

System	Examples	Strength	Gap Port Command fills
Terminal operating system (TOS)	Vendors such as Navis, named in smart-port market research	System of record for berth, yard and gate planning and container tracking	Planning is screen by screen and people connect the dots. We add one live map and an AI that explains and proposes; we read from the TOS and never replace it
Automated terminal control	Qingdao New Qianwan, Shanghai Yangshan Phase IV, Tianjin	Automated cranes and vehicles, remote control, AI scheduling	Tied to automated hardware and new-build or heavy retrofit; we target operating ports with a mixed fleet
MMC's own programme with NIZE	Digital Blueprint Roadmap, group dashboard, digital twin pilot (The Star)	The foundation MMC has chosen	A neighbour, not a rival: we show an AI decision layer and a role-based front end on top. Judges may see overlap, so pitch it as a complement
Operational technology partnerships	Siemens at PTP (MMC Ports)	Equipment and control-system integration pilots	Same relationship: we consume their data and add decisions
AIS vessel trackers	Open-source apps such as ship-tracker and risk platforms such as Meridian	Ships at sea, with a click-to-inspect panel	Stop at the port boundary: no yard, vehicles, workers or decisions

System	Examples	Strength	Gap Port Command fills
Port community systems	Malaysia Maritime Single Window	Clearance, documents and payments across agencies	Not operational control; we show its clearance status on the map

What judges should see in 30 seconds

1. One map from the approach channel to the gate, including people.
2. An AI that explains and proposes, with a human approving.
3. Role-based portals that include safety, compliance and cyber, not only operations.
4. A layer that reads from existing systems instead of replacing them.

The prototype must never claim to be a terminal operating system or a crane controller.

Design language

The port is drawn as a printed circuit board: a light neutral board, black traces for roads, and components for berths, cranes and yard blocks. It should read as technical drawing, not as a game or a sky-blue dashboard.

Circuit-board rules for the map

- Roads are black traces: 4 px for main arteries, 2.5 px for secondary roads, 1.5 px for yard lanes.
- Traces turn only at 90 or 45 degrees, with chamfered corners, as on a real board.
- Junctions and gates get small ring pads.
- Each berth is drawn like a chip footprint: a rectangle on the quay edge with short pin ticks, labelled B1 to B4 in monospace.
- Yard blocks are grids of small squares, one per container slot; utilisation can show as a flat tint.
- Water is a quiet grey-blue with faint contour lines, with no gradients or waves.
- A very light 8 px dot grid sits under everything.
- When the AI assigns a job, a thin pulse travels once along the route trace, like a signal. Use it sparingly.

Vehicles, ships and the 3D illusion

- Keep the world strictly 2D and top down, and fake height with a flat extrusion: the top face plus a darker side face offset 3 to 6 px.

- One light source for everything, top left, with a soft shadow offset down and right at 20% to 30% opacity.
- Container stacks show height as extruded layers; cranes show legs, boom and a shadow that moves with the boom.
- Vehicles and ships rotate to follow their path and ease through turns; they never snap.
- Use a fixed sprite set: prime mover, AGV, reach stacker, yard crane, quay crane, container ship, feeder, tug. Draw them as vectors or pre-render at 2x.

Tokens (proposed; adjust by eye)

Token	Value	Use
Board	#F2F3EF	Map background
Land block	#E4E6DF	Yard, apron, buildings
Water	#C8D1D6	Sea and channel
Trace	#15171A	Roads and borders
Ink secondary	#5B6068	Labels and secondary text
People blue	#2563EB	Worker pins only
AI teal	#0F766E	Copilot and proposals
Warning amber	#B45309	Advisory and warning
Critical red	#B91C1C	Critical alerts and breaches
OK green	#15803D	Normal and resolved

Blue is reserved for people, so the screen is not blue overall. Status is never colour alone: pair it with an icon or a shape.

Shape and type

- One radius scale: 4 px for chips and inputs, 8 px for cards and panels, 12 px for drawers and modals. Map elements stay square or chamfered to keep the board feel.
- Inter for interface text and a monospace such as JetBrains Mono for IDs, coordinates and container numbers; both are open source. Use tabular numerals for live counters.
- Type sizes 12, 14, 16, 20 and 28 px, and nothing smaller than 12 px.

Do not

- No neon, glow, gradients or scan-line effects.
- No game-style frames, health-bar HUDs, sound effects or particle bursts.

- No cartoon vehicles, emoji or mixed icon styles.
- No tilting or rotating the whole map in 3D.
- No more than one animated element drawing attention at a time.

Open question: you mentioned a specific rounding style from a piece of research. If you meant continuous corners (the squircle used in Apple interfaces) rather than a plain radius, say so and the shape rules change.

Portals and screens

One shared simulated state feeds ten portals; build Terminal Control first because it carries the demo.

Shell (the same on every screen)

- Top bar: terminal selector (Group, then Selat Terminal and the five MMC ports as illustrative tiles), live clock in Malaysia time, role switcher, "Simulated data" badge, alert bell.
- Left rail, collapsible: the overview panel with KPI tiles (vessels at berth, vessels waiting, quay crane moves per hour, gate queue, workers on shift, open alerts), the live alert feed and the AI button. It collapses to icons so the map can fill the screen.
- Centre: the portal's main view; for Terminal Control it is the circuit-board map.
- Right: an inspector that slides in when anything is clicked and closes with Esc.
- Bottom: a thin status strip with scenario clock and speed (live, 2x, 5x). Demo controls stay hidden behind the D key.

Portal	Layout and key widgets	Demo behaviour	Priority
Terminal Control	Full-screen map, layer toggles, KPI tiles, alert feed, inspector	Scenes 1 to 5 run here	Must
AI Copilot	Drawer opened with the / key: thread, prompt chips, proposal cards	Answers 12 scripted prompts; approves proposals	Must
Safety and Workforce	Zone map, live breach list, contractor roster by employer, incident log with rule tags, certificate expiry list	Scene 4 writes the incident log	Must

Portal	Layout and key widgets	Demo behaviour	Priority
Group Dashboard	Five terminal tiles with simulated KPIs, 2025 TEU totals as static reference, drill-down to a terminal	Scene 6	Must
Vessel and Berth Planning	Berth timeline (Gantt) with vessel bars, ETA drift markers, delay predictions	Dragging a bar re-plans; the AI proposal does the same	Should
Yard and Equipment	Block heat map, equipment list with utilisation, idle and fault states, dispatch queue	The scene 5 proposal moves two prime movers	Should
Compliance and Audit	Rule cards (safety, cyber, data protection, single window) with status, evidence links and an export button	Export produces a mock PDF named with the date	Should
Gate and Landside	Truck queue, appointment slots, gate throughput chart	Queue grows and clears	Could
Energy and Sustainability	Electric equipment state of charge, charging bays, emissions per vessel call	Slow trending numbers	Could
Cyber and OT Security	Asset list, alert stream, incident notification clock	One injected low-severity alert	Could
Shipping Line Portal	Customer view: their vessels, container status, berth slot	Search by container ID	Could

The left overview panel is the hero area: show it open for the first scene, then collapse it when the map needs the full width.

Map behaviour and interactions

The map behaves like a real map, with free pan and zoom, but draws a fictional circuit-board port so the audience reads it instantly.

Zoom levels

Level	What shows	Detail
Overview	Whole terminal and approach channel	Vessels, berths, main traces, worker clusters, alert markers
Berth	Two berths in view	Crane booms, mooring lines, vehicles, individual worker pins
Yard	One yard block	Container slots with extruded stacks, reach stackers, slot labels

Switch detail by zoom level (labels, sprites, pins), animate zoom over 300 to 400 ms with easing, and fly to any clicked alert or search result.

Layers (toggles in a floating card): vessels, vehicles, workers, containers, zones, utilisation heat, route pulses.

Worker pins. Draw each as the familiar live-location dot: a blue core, a white ring and a faint pulsing halo. Update about once a second with eased movement, and cluster at overview zoom. Turn amber near a zone edge and red inside an exclusion zone, with a small icon so colour is not the only signal.

Click behaviour. Clicking anything selects it with a thin outline, centres it gently and opens the inspector on the right. Hover shows a short tooltip with name and status.

Entity	Inspector shows
Vessel	Name, type, flag, fictional IMO number, origin to destination, ETA or ATA, berth, discharge and load counts, status (approaching, pilotage, docking, at berth, departing), predicted delay, single-window clearance status, Ask AI button
Vehicle	ID, type, driver or autonomous, current task, from and to, container carried, speed, battery for electric units, status
Worker	Fictional name, role, employer, contractor flag, shift, current task, zone, safety status, certificate expiry; zone-level position by default, exact position only in the supervisor role
Container	Fictional ISO-style ID, size, type, vessel, yard slot (block, row, bay, tier), status, last move, hold flag

Entity	Inspector shows
Crane or berth	ID, status, current move, moves per hour, next job, certificate expiry, exclusion zone on or off

Zones. Draw exclusion zones around active cranes as hatched polygons, with walkways and restricted lanes in a lighter tone. A zone is active only while its crane is working.

Alerts. Four levels: info, advisory, warning, critical. Each appears as a toast at top right, a row in the feed and a marker on the map; clicking a row flies to the location. Only critical alerts pulse.

Keyboard. The / key opens the copilot, Esc closes the inspector or drawer, 1 to 4 switch portals, D shows demo controls, and Shift with 1 to 6 jumps to a scene.

Performance target. 60 frames per second with 3 vessels, about 40 vehicles, about 60 pins and 2,000 container slots on the presenting laptop. Draw on canvas or WebGL, not thousands of DOM nodes.

Demo scenes

Six scripted scenes run on a fixed timeline so the stage demo is identical every time; each starts from its shortcut and can be reset. All names are fictional.

Vessel	Type	Route	Role in the demo
MV Kedah Pearl	Container ship	Shanghai to Selat Terminal, then Colombo	Docks at Berth 2 in scene 3
MV Orient Gale	Container ship	Singapore to Selat Terminal	Approaching; predicted late in scene 5
MV Langkawi Dawn	Feeder	Selat Terminal to Penang	Departs Berth 4 during scene 1

Scene	Pitch time	What happens	Must be visible
1 Calm overview	0:00	Vessels, vehicles and pins move; KPI tiles tick; Langkawi Dawn leaves Berth 4	Zoom in and out; the left panel open
2 Inspect	1:00	Click Orient Gale, a worker pin and a container in turn	Three different inspectors

Scene	Pitch time	What happens	Must be visible
3 Docking	1:45	Kedah Pearl enters the docking zone, tugs attach, mooring completes, cranes swing, prime movers start their cycles	Docking guide brackets, move counter rising
4 Safety	2:30	A contractor walks into Crane 3's exclusion zone; the pin turns red; the system holds Crane 3 and proposes a lane change; the supervisor approves; the incident is logged with a safety-law tag	Red pin, alert, approval, log entry
5 Decide	3:15	Copilot is asked why Berth 2 is behind; it cites the Crane 3 hold and two missing prime movers and proposes moving two prime movers from Yard C and swapping a crane; after approval vehicles reroute and Orient Gale's predicted delay shrinks	Proposal card, reroute animation, delay figure changing (all scripted)
6 Scale	4:00	Group Dashboard with five tiles; click one to drill down	Illustrative-data badge on the tiles

Set the scene 5 figures yourself, for example a delay shrinking from 24 to 13 minutes. They are simulated and must say so on screen.

On the Group Dashboard, use MMC's port names only with an "Illustrative data" badge. The only real numbers allowed there are the public 2025 TEU totals listed in the pitch document; every other KPI is simulated.

Build a scenario runner: a list of timed events keyed to scenario time, a reset to the start state, and a seed for the ambient motion so every run looks the same.

AI copilot specification

The copilot answers from the simulated state, never from free text, so every answer can be checked on the map and replays identically on stage.

Interface. A right-hand drawer opened with the / key or the teal AI button, holding a thread, prompt chips and proposal cards. Entities named in an answer are chips: hovering one highlights it on the map and clicking flies to it.

Answer format. Lead with the answer in one sentence, add two or three evidence lines, then a proposal card if action is possible. Every answer carries a small "Simulated data" tag.

Proposal card. Title, reason, expected effect (for example minutes saved), confidence, and three buttons: Approve, Modify, Dismiss. Approving runs the change in the simulation and writes a log entry with the approver's role and the time.

Autonomy control. A three-way switch labelled Advisory, Approve to act and Auto for low-risk changes shows the governance idea. In the demo only Approve to act works; the other two show a short explanation.

How it stays grounded. The copilot calls a small set of functions on the state store, for example getVesselStatus, getBerthDelay, getEquipmentAvailability, getAlerts, listWorkersInZone and proposeReallocation. It has two modes:

1. Scripted (the default on stage): the question is matched to one of the 12 prompts below and a fixed answer is built from live state values.
2. Live model (optional): any language model with tool calling can call the same functions. If the network or the model fails, fall back to scripted mode silently.

The 12 prompts to support

Prompt	Behaviour
What is happening at Berth 2?	Status, moves per hour, active crane and vessel
Why is Berth 2 behind schedule?	Cites the Crane 3 hold and the prime mover shortage; offers a proposal
Which vessel will be late?	Orient Gale, with the predicted delay
Who is in a crane exclusion zone?	Lists pins inside zones, or none
Show last night's safety incidents	Opens the Safety incident log filtered to the last shift
What should we change to recover the delay?	The scene 5 proposal card
What if Orient Gale arrives 20 minutes late?	Re-plans on screen and shows the changed berth timeline

Prompt	Behaviour
Summarise this shift	A short paragraph with counts of moves, alerts and approvals
Is Crane 3's certificate valid?	Expiry date and status from the equipment record
How much energy have electric prime movers used today?	Simulated kWh and share of moves
Prepare this week's audit report	Opens a mock report in the Compliance portal
Compare all five ports	Switches to the Group Dashboard with a short ranking

Rules the copilot always follows

- It never moves equipment without a human approval.
- It states its confidence and offers a Why link that shows the data it used.
- Every proposal and approval is logged with a time and a role.
- It says it does not have the answer in this simulation instead of inventing one.

Optional extras, in order: voice input through the browser speech API (some browsers need internet for it), a typing animation tuned to feel quick, and a what-if slider on vessel delay.

Simulated data and event engine

A fixed-step simulation loop updates one shared state store, and every screen, the rule engine and the copilot read from that store, so the front end needs no real backend.

Entities

Entity	Key fields	Demo count	Update rate
Vessel	id, name, fictional IMO, type, length, origin, destination, ETA, status, position, heading, speed, berth, discharge and load counts	3	1 per second
Berth	id, name, quay polygon, crane ids, occupant	4	On change

Entity	Key fields	Demo count	Update rate
Crane	id, type (quay or yard), location, status, current move, moves per hour, certificate expiry, exclusion zone	6 quay, 8 yard	1 per second
Vehicle	id, type, driver or autonomous, position, heading, speed, task, container, battery	About 40	5 per second
Worker	id, fictional name, role, employer, contractor flag, shift, position, zone, status, certificate expiry	About 60	1 per second
Container	id, size, type, vessel, yard slot (block, row, bay, tier), status, hold flag	About 2,000, few moving	On change
Zone	id, polygon, type (exclusion, walkway, restricted), active-when rule	About 12	On change
Alert	id, severity, type, entity references, position, time, status, rule tag	A few	On event
Proposal	id, title, reason, effect, confidence, actions, status	1 to 3	On event
Log entry	time, actor, action, entity references, rule tag	Grows over time	On event

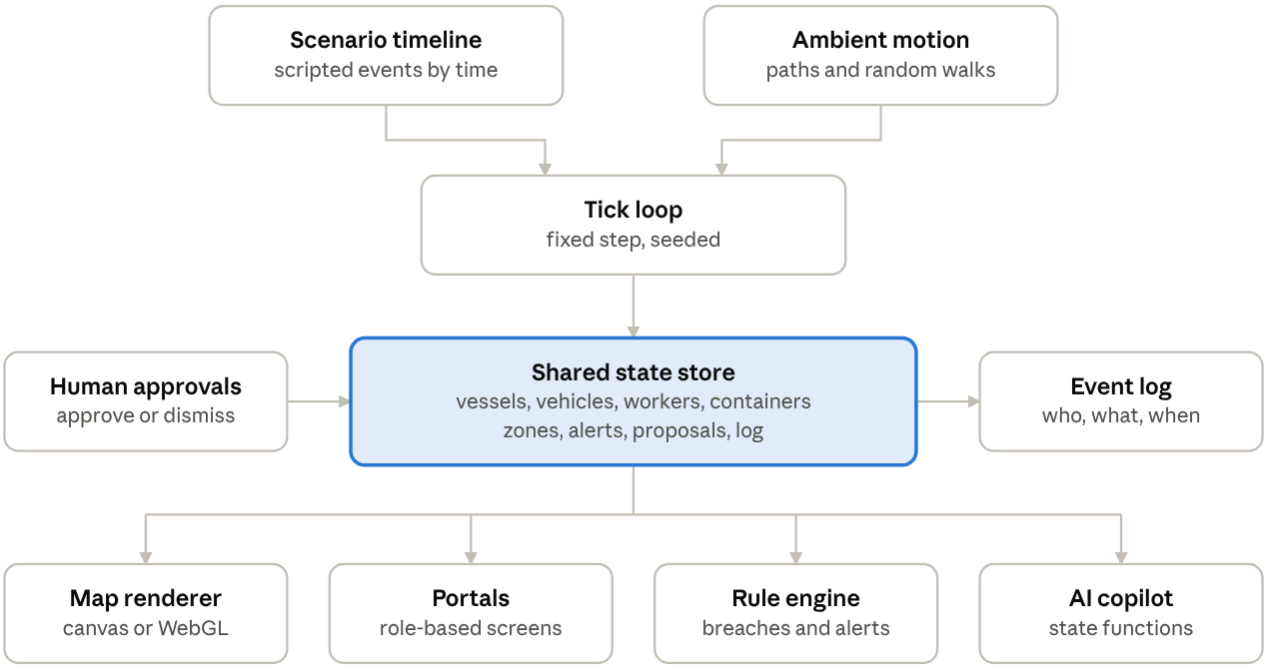
Event types. VESSEL_ARRIVE, DOCK_START, DOCK_COMPLETE, CRANE_MOVE, ZONE_BREACH, CRANE_HOLD, PROPOSAL_CREATED, PROPOSAL_APPROVED, ROUTE_CHANGE, ALERT_RESOLVED.

Rules. A zone breach fires when a worker position falls inside an active exclusion polygon (a point-in-polygon test). Delay prediction can be a simple formula on queue length and crane availability; it only has to agree with the scripted scenes.

Motion. Vessels follow smooth curves into berth, vehicles follow pre-defined polylines with eased turns, and workers walk randomly inside walkable zones. Run the simulation at a fixed rate (for example 10 ticks per second, ideally in a web worker) and interpolate for rendering at 60 frames per second. Seed the random generator so replays match.

The drawing below shows how the parts connect.

One tick loop feeds a single state store that every screen reads



simulation data flow, 12 parts

Alerts raised by the rule engine and proposals approved by a person are written back to the store as events, so the log and every screen stay in step.

Open-source building blocks

Use these directly; the table says what each is for and its licence. A licence marked with an asterisk comes from general knowledge, not from a page we read, so confirm it on the repository before shipping.

Need	Use	Licence	How to use it here
Map engine	MapLibre GL JS	3-Clause BSD	Start from an empty style and draw the port as custom GeoJSON layers; you keep real-map pan, zoom and fly-to
Geometry maths	Turf.js (@turf/turf), listed on the same page	MIT	Point-in-polygon for zone breaches, distances, positions along a path

Need	Use	Licence	How to use it here
Extra layers	deck.gl, listed on the same page	MIT*	Optional: animated trips and extruded shapes if MapLibre layers are not enough
Route simulation	Eclipse SUMO	Eclipse Public License 2.0	Optional: generate vehicle routes offline and replay them; a hand-made path set is simpler
Live AIS for an outside-the-port view	AISStream.io, used by ship-tracker and BoatOS	Service terms; free API key	Optional and risky on stage: record a few minutes of messages and replay them offline
Reference app: vessel detail panel	ship-tracker (React, Leaflet)	Not stated in what we read; check	Read it for the click-to-inspect pattern and vessel type colouring; do not copy code until the licence is confirmed
Reference app: map plus risk dashboard	Meridian (React, Vite, MapLibre, Recharts)	Apache 2.0 (badge on the repository)	A good model for command-centre layout and KPI cards; code may be reused under its licence terms
App shell	React, Vite, Tailwind CSS	MIT*	Standard front-end stack; any equivalent works
State	Zustand or similar	MIT*	One shared store for all portals
Charts	Recharts or similar	MIT*	Berth timeline, KPI sparklines, throughput
Fonts	Inter, JetBrains Mono	SIL Open Font Licence*	Self-host the font files so the demo works offline

Need	Use	Licence	How to use it here
Real port outline, only if wanted	OpenStreetMap	ODbL*	Trace a real layout; attribution is required

Licence warning. The open-source intelligence dashboard Skopia is under AGPL v3, which requires you to release your own source under the same licence if you build on it. Read it for ideas only, unless you accept that.

Offline rule: self-host every library, font, map tile and icon. Nothing on the stage laptop should load from the internet.

Build plan and acceptance checklist

Build in this order so a convincing demo exists early and every later step only adds to it.

1. Map shell: empty MapLibre style, circuit-board layers (traces, pads, berths, yard blocks, water), left rail, right inspector.
2. Simulation core: tick loop, state store, ambient motion for vessels, vehicles and pins along fixed paths.
3. Click-to-inspect for the five entity types, then layer toggles and zoom levels.
4. Scenario runner and scenes 1 to 5, with alerts, zones and the approval flow.
5. Copilot drawer with the 12 scripted prompts and proposal cards.
6. Group Dashboard and Safety and Workforce, then the Should portals.
7. Polish: sprites and shadows, transitions, projector test, recorded backup video.
8. Could items only if time remains.

Tier	Items
Must	Terminal Control map and inspectors, scenes 1 to 5, AI Copilot, Safety and Workforce, Group Dashboard, simulated-data badge, offline run
Should	Vessel and Berth Planning, Yard and Equipment, Compliance and Audit, scene 6, projector check
Could	Gate, Energy, Cyber and Shipping Line portals, voice input, what-if slider, replayed AIS context map, dark theme

Acceptance checklist (test the day before)

- [ ] Opens full screen at 1920 by 1080 and at 1366 by 768 with no scrollbars.

- [ ] Runs with Wi-Fi switched off.
- [ ] Holds 60 frames per second with the full demo population on the presenting laptop.
- [ ] Scenes 1 to 6 each start from their shortcut and reset cleanly.
- [ ] Clicking a vessel, vehicle, worker, container and crane opens the right inspector.
- [ ] The zone breach turns the pin red, raises the alert, holds the crane and writes the log entry.
- [ ] The copilot answers all 12 prompts and the approved proposal changes the simulation.
- [ ] A "Simulated data" badge shows on every screen.
- [ ] No real names, faces or real operational figures appear.
- [ ] Legible on a bright projector; the light board does not wash out.
- [ ] A full screen recording of the demo exists as a backup.

Open questions

- The competition's time limit, and whether judges expect any live data.
- The framework the teammate prefers; the stack above is a suggestion, not a requirement.
- The rounding style mentioned in the design section.

Sources

Research for this specification was done on 28 October 2026 from the pages below. The MMC briefing, benchmarks and regulation sources are in the pitch document. Re-check every licence on its repository before shipping.

- Coherent Market Insights: smart ports market
- The Star: MMC Port and NIZE blueprint
- MMC Ports press releases
- Cambridge Intelligence: libraries and licences list
- Eclipse SUMO repository
- ship-tracker repository
- Meridian repository
- BoatOS repository
- Skopia repository